

SOLOMON AHEDOR

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EDUCATION

Kwame Nkrumah University of Science and Technology (KNUST) – Kumasi, Ghana

BSc Metallurgical Engineering – Expected Completion: 2026

Relevant Coursework: Extractive Metallurgy, Mineral Processing, Data Science, Physical Metallurgy, Materials Characterization, Machine Learning, Artificial Intelligence, Material Informatics, Computational Materials.

TECHNICAL SKILLS

- **Metallurgical Processes:** Refractory Ore Processing, Bacterial Oxidation (BIOX), Froth Flotation, Carbon-in-Leach (CIL), Elution Circuits.
- **Laboratory Instrumentation:** LECO (S832 DR/S632) for Sulfur and Carbon analysis, Atomic Absorption Spectroscopy (AAS), Particle Size Analyzers, Optical Microscopy.
- **Computational Modeling & Data:** Python (Pandas, NumPy, Matplotlib), Model Predictive Control, Machine Learning (Scikit-learn).
- **Software & Dev:** Microsoft Office Suite, Python, yEd Graph Editor, Design Expert, MATLAB, Ansys Granta Selector, Github.

PROFESSIONAL EXPERIENCE

AngloGold Ashanti (Obuasi Mine), Obuasi, Ghana

Metallurgy & Process Engineering Intern, Nov 2025 – Dec 2025

- **Process Circuit:** Completed technical rotation through the full refractory gold processing circuit and Monitored BIOX bacterial activity and performed ferrous/ferric ion determinations to support optimization of sulfide oxidation rates and CIL adsorption efficiency.
- **Laboratory Analysis:** Performed extended flotation tests and Particle Size Analysis (PSA) in the Met-lab to evaluate recovery efficiency and grind kinetics, providing data to refine process settings.

- **Analytical Chemistry:** Assisted with LECO instrumentation (S832 DR and S632) for total sulfur, carbon, and carbonate analysis, and utilized AAS for detecting gold and arsenic concentrations in solution.

RESEARCH & ACADEMIC PROJECTS

Department of Materials & Metallurgical Engineering, KNUST, Kumasi, Ghana

Undergraduate Researcher – Process Dynamics & Optimization, Sept 2025 – Present

- **Project:** Optimizing Froth Flotation Performance through Dynamic Modelling.
- **Modelling:** Trained an XGBoost models to predict grade and recovery on small and large datasets, achieving R^2 scores of 65% to 95%.
- **Performance targets:** Used Python libraries (Pandas, NumPy) to improve grade and recovery targets.

MetaForge: ML-Driven High Entropy Alloy Discovery Platform, KNUST, Kumasi, Ghana

- **Development:** Built an end-to-end HEA discovery pipeline combining 132 Matminer Magpie descriptors, Random Forest regression, genetic algorithm inverse design, and CHGNet structural relaxation.
- **Performance:** Achieved density RMSE of 0.073 g/cm³ ($R^2 = 0.99$) and strength RMSE of 0.539 GPa ($R^2 = 0.95$); top candidate composition W_{0.10}Mo_{0.40}Ta_{0.05}Nb_{0.05}V_{0.40}.
- **Publication:** Deployed the platform publicly at <https://metaforge-web.onrender.com/> and open-sourced the codebase on GitHub.

LEADERSHIP & AWARDS

Executive Head & R&D, Materials Innovation Hub (MIH) | KNUST | 2026 Term

- Oversee policy, events, peer mentorship, and student R&D team across active research projects in biopolymers.

Citation Recipient, Hult Prize National Competition (2025)

2nd Runner-Up, Hult Prize KNUST Campus Edition (2025)

Zonal Participant, Mastercard-Kosmos Innovation Center Competition - awarded \$300 for market research.

LANGUAGES

English (Fluent) | Twi (Fluent) | German (Basic)